

PVsyst - Simulation report

Grid-Connected System

Project: caso2-guaira-pr

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 3000 Wp

Guaira_PR - Brazil



PVsyst V8.1.5

VCO, Simulation date:
04/09/26 14:09
with V8.1.5

Project summary

Geographical Site

Guaira_PR
Brazil

Situation

Latitude -24.0800 °(S)
Longitude -54.2558 °(W)
Altitude 270 m
Time zone UTC-3

Project settings

Albedo 0.20

Weather data

Guaira_PR
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

Orientation #1 Horizontal plane

System information

PV Array

Nb. of modules
Pnom total

No 3D scene defined, no shadings

Near Shadings

no Shadings

Inverters

Nb. of units
Total power
Pnom ratio

User's needs

Unlimited load (grid)

5 units
3000 Wp

1 unit
2.5 kWac
1.20

Results summary

Simulation step Hourly

Produced Energy 4574.1 kWh/year Specific production 1525 kWh/kWp/year Perf. Ratio PR 84.68 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Horizontal plane

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic
Model JKM-600N-72HL4-V
(Original PVsyst database)
Unit Nom. Power 600 Wp
Number of PV modules 5 units
Nominal (STC) 3000 Wp
Modules 1 unit x 5 In series
At operating cond. (50°C)
Pmpp 2783 Wp
U mpp 204 V
I mpp 14 A

Total PV power

Nominal (STC) 3.00 kWp
Total 5 modules
Module area 12.9 m²
Cell area 11.9 m²

Inverter

Manufacturer Generic
Model MIC 2500TL-X
(Original PVsyst database)
Unit Nom. Power 2.50 kWac
Number of inverters 1 unit
Total power 2.5 kWac
Operating voltage 65-550 V
Pnom ratio (DC:AC) 1.20

Total inverter power

Total power 2.5 kWac
Number of inverters 1 unit
Pnom ratio 1.20

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 245 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

Simulation step

Hourly

System Production

Produced Energy 4574.1 kWh/year

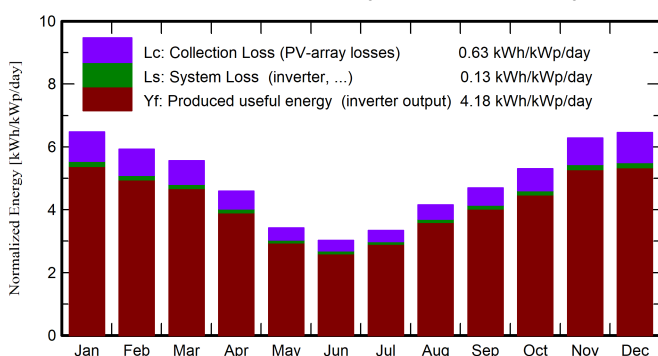
Specific production

1525 kWh/kWp/year

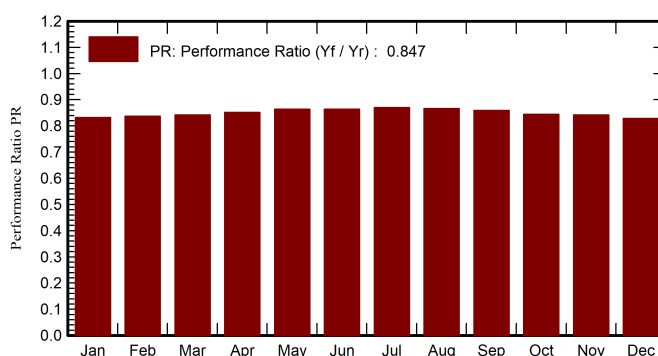
Perf. Ratio PR

84.68 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	200.8	76.04	27.06	200.8	196.2	516.3	501.4	0.832
February	166.0	65.63	26.84	165.9	162.0	428.8	416.4	0.837
March	172.4	59.13	26.23	172.3	167.7	448.2	435.4	0.842
April	138.0	44.86	23.85	137.9	133.5	362.7	352.2	0.851
May	106.2	39.13	19.41	106.2	101.8	283.6	275.1	0.864
June	90.7	34.76	17.93	90.7	86.6	242.7	235.1	0.864
July	103.7	35.42	17.40	103.7	99.4	279.0	270.6	0.870
August	129.0	40.43	19.14	128.9	124.4	344.9	335.0	0.866
September	141.0	47.66	21.85	140.9	137.1	374.3	363.3	0.859
October	164.5	62.19	24.70	164.4	160.4	428.9	416.3	0.844
November	188.7	63.79	25.41	188.6	184.3	490.0	476.1	0.841
December	200.3	76.47	26.65	200.1	195.4	512.0	497.4	0.828
Year	1801.4	645.52	23.02	1800.5	1748.8	4711.3	4574.1	0.847

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

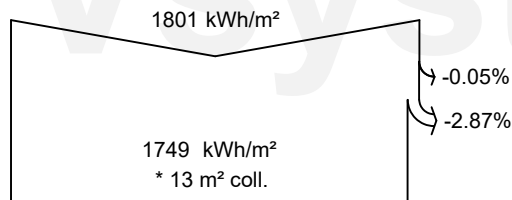
PR Performance Ratio



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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

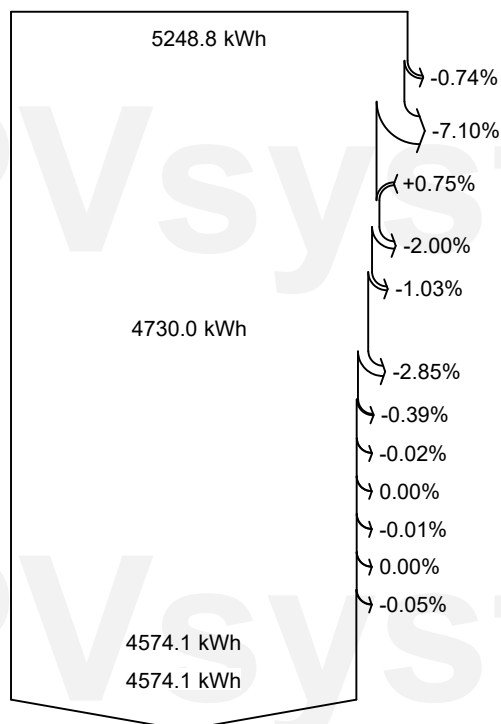
Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

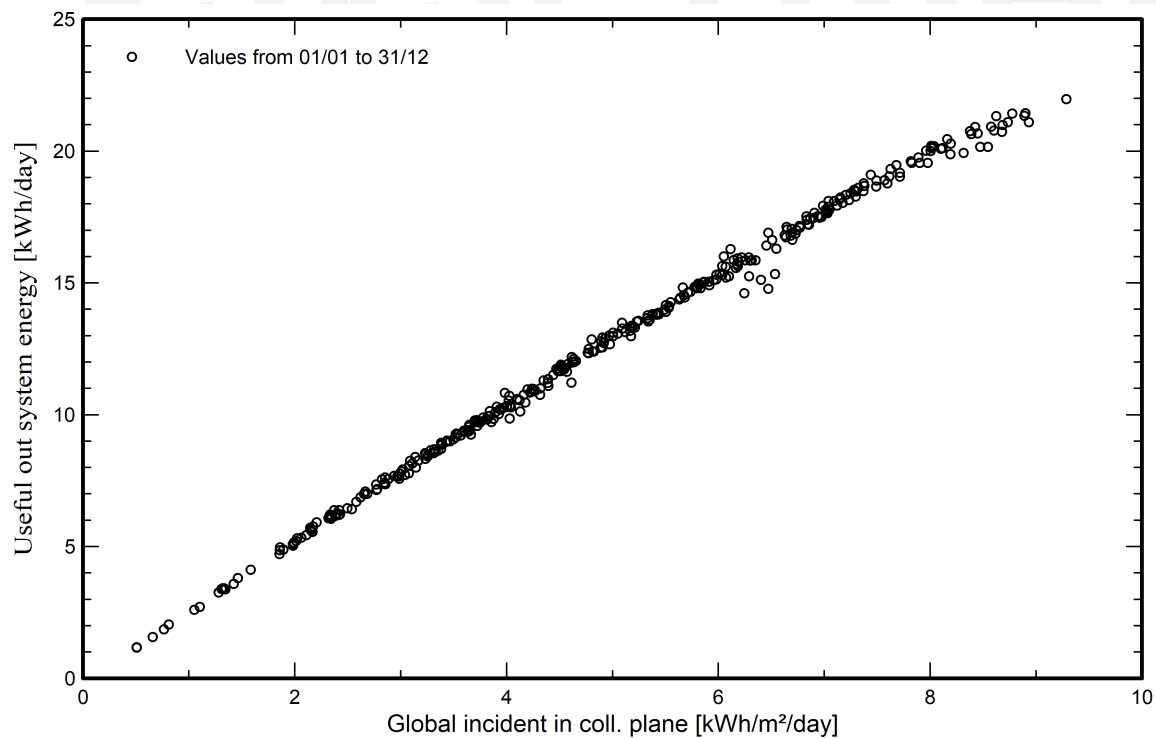




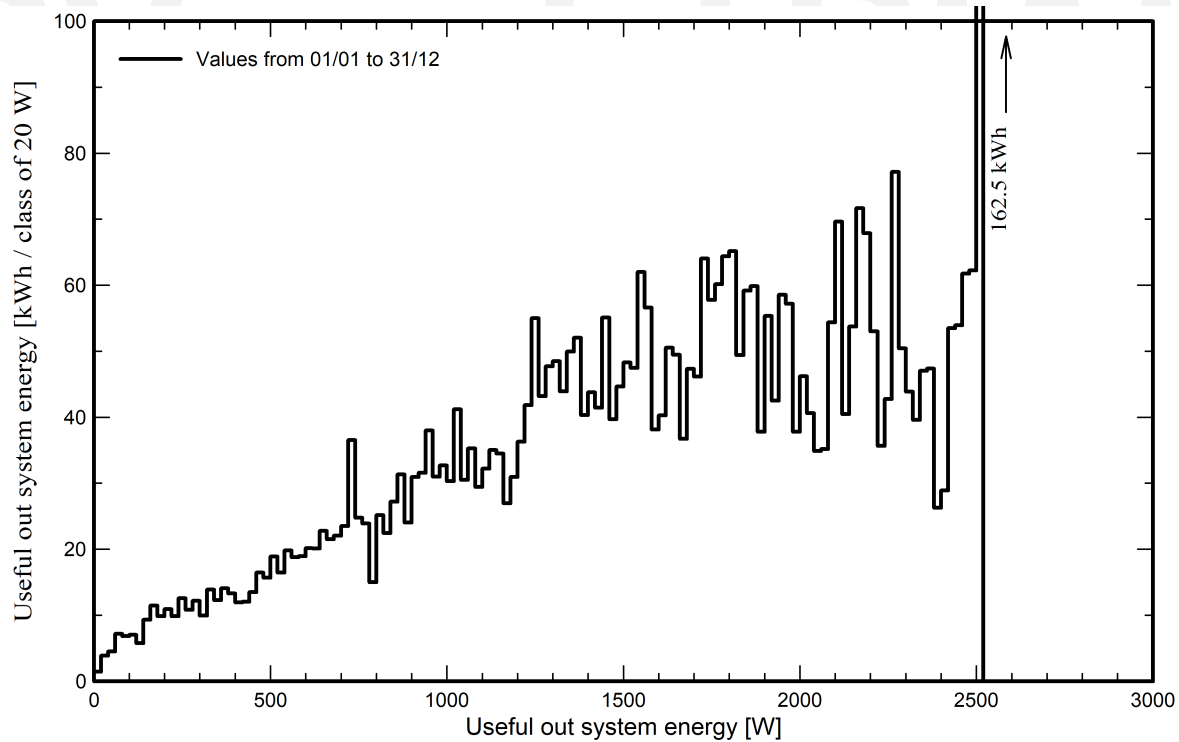
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

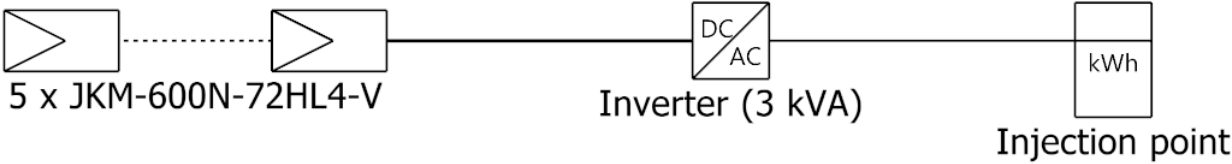




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Single-line diagram



PV module	JKM-600N-72HL4-V
Inverter	MIC 2500TL-X
String	5 x JKM-600N-72HL4-V

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VC0 : New simulation variant

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